



Report

SHA

Name /Mohammed Talal Alobaysi

Input Message

Message:

"MohammedTalal443170285SHA1is160bitHashFunction_DesignedByNSAin1995_
ProducesA40DigitHexOutput_UsesMessageBlocks512Bits_InitialHashValueI
sA1A2A3A4A5_CommonInGitAndTLS10_MD5base"

Length: 179 characters

Initial Hash Values

H0 = 0110 0111 0100 0101 0010 0011 0000 0001 (0x67452301)

H1 = 1110 1111 1100 1101 1010 1011 1000 1001 (0xefcdab89)

H2 = 1001 1000 1011 1010 1101 1100 1111 1110 (0x98badcfe)

H3 = 0001 0000 0011 0010 0101 0100 0111 0110 (0x10325476)

H4 = 1100 0011 1101 0010 1110 0001 1111 0000 (0xc3d2e1f0)

Message Preprocessing

SHA-1 message preprocessing involves padding the message to a multiple of 512 bits.

Original Message:

"MohammedTalal443170285SHA1is160bitHashFunction_DesignedByNSAin1995_
ProducesA40DigitHexOutput_UsesMessageBlocks512Bits_InitialHashValueI
sA1A2A3A4A5_CommonInGitAndTLS10_MD5base"

Message in binary:

```
01000001 01100010 01100100 01110101 01101100 01101100 01100001
01101000 01001000 01100001 01101101 01111010 01100001 01101000
01001001 01100010 01110010 01100001 01101000 01101001 01101101
01100011 01101111 01110011 00110100 00110111 00110001 01010011
01001000 01000001 00110001 01101001 01110011 00110001 00110110
00110000 01100010 01101001 01110100 01001000 01100001 01110011
01101000 01000110 01110101 01101110 01100011 01110100 01101001
01101111 01101110 01011111 01000100 01100101 01110011 01101001
01100111 01101110 01100101 01100100 01000010 01111001 01001110
01010011 01000001 01101001 01101110 00110001 00111001 00111001
00110101 01011111 01010000 01110010 01101111 01100100 01110101
01100011 01100101 01110011 01000001 00110100 00110000 01000100
01101001 01100111 01101001 01110100 01001000 01100101 01111000
01001111 01110101 01110100 01110000 01110101 01110100 01011111
01010101 01110011 01100101 01110011 01001101 01100101 01110011
01110011 01100001 01100111 01100101 01000010 01101100 01101111
01100011 01101011 01110011 00110101 00110001 00110010 01000010
01101001 01110100 01110011 01011111 01001001 01101110 01101001
01110100 01101001 01100001 01101100 01001000 01100001 01110011
01101000 01010110 01100001 01101100 01110101 01100101 01001001
01110011 01000001 00110001 01000001 00110010 01000001 00110011
01000001 00110100 01000001 00110101 01011111 01000011 01101111
01101101 01101101 01101111 01101110 01001001 01101110 01000111
01101001 01110100 01000001 01101110 01100100 01010100 01001100
01010011 00110001 00110000 01011111 01001101 01000100 00110101
```

01100010 01100001 01110011 01100101

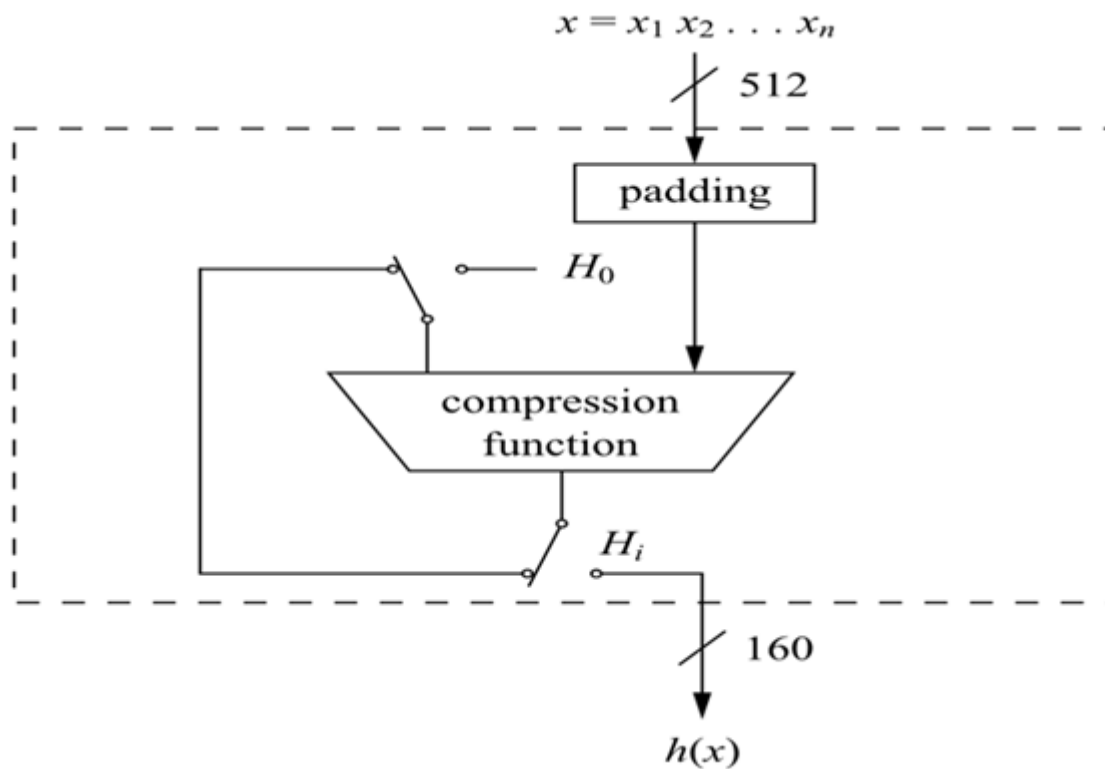
Length: 179 bytes (1432 bits)

After padding and reserving 64 bits:

01000001 01100010 01100100 01110101 01101100 01101100 01100001
01101000 01001000 01100001 01101101 01111010 01100001 01101000
01001001 01100010 01110010 01100001 01101000 01101001 01101101
01100011 01101111 01110011 00110100 00110111 00110001 01010011
01001000 01000001 00110001 01101001 01110011 00110001 00110110
00110000 01100010 01101001 01110100 01001000 01100001 01110011
01101000 01000110 01110101 01101110 01100011 01110100 01101001
01101111 01101110 01011111 01000100 01100101 01110011 01101001
01100111 01101110 01100101 01100100 01000010 01111001 01001110
01010011 01000001 01101001 01101110 00110001 00111001 00111001
00110101 01011111 01010000 01110010 01101111 01100100 01110101
01100011 01100101 01110011 01000001 00110100 00110000 01000100
01101001 01100111 01101001 01110100 01001000 01100101 01111000
01001111 01110101 01110100 01110000 01110101 01110100 01011111
01010101 01110011 01100101 01110011 01001101 01100101 01110011
01110011 01100001 01100111 01100101 01000010 01101100 01101111
01100011 01101011 01110011 00110101 00110001 00110010 01000010
01101001 01110100 01110011 01011111 01001001 01101110 01101001
01110100 01101001 01100001 01101100 01001000 01100001 01110011
01101000 01010110 01100001 01101100 01110101 01100101 01001001
01110011 01000001 00110001 01000001 00110010 01000001 00110011
01000001 00110100 01000001 00110101 01011111 01000011 01101111
01101101 01101101 01101111 01101110 01001001 01101110 01000111
01101001 01110100 01000001 01101110 01100100 01010100 01001100
01010011 00110001 00110000 01011111 01001101 01000100 00110101
01100010 01100001 01110011 01100101 10000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000101 10011000

Final padded message length: 192 bytes (1536 bits)

Processing Chunk 1 of 3



Message Schedule Array (W) for Chunk 1

$W[0] = 0100\ 0001\ 0110\ 0010\ 0110\ 0100\ 0111\ 0101\ (0x41626475)$

$W[1] = 0110\ 1100\ 0110\ 1100\ 0110\ 0001\ 0110\ 1000\ (0x6c6c6168)$

Initial Hash Values for Chunk 1

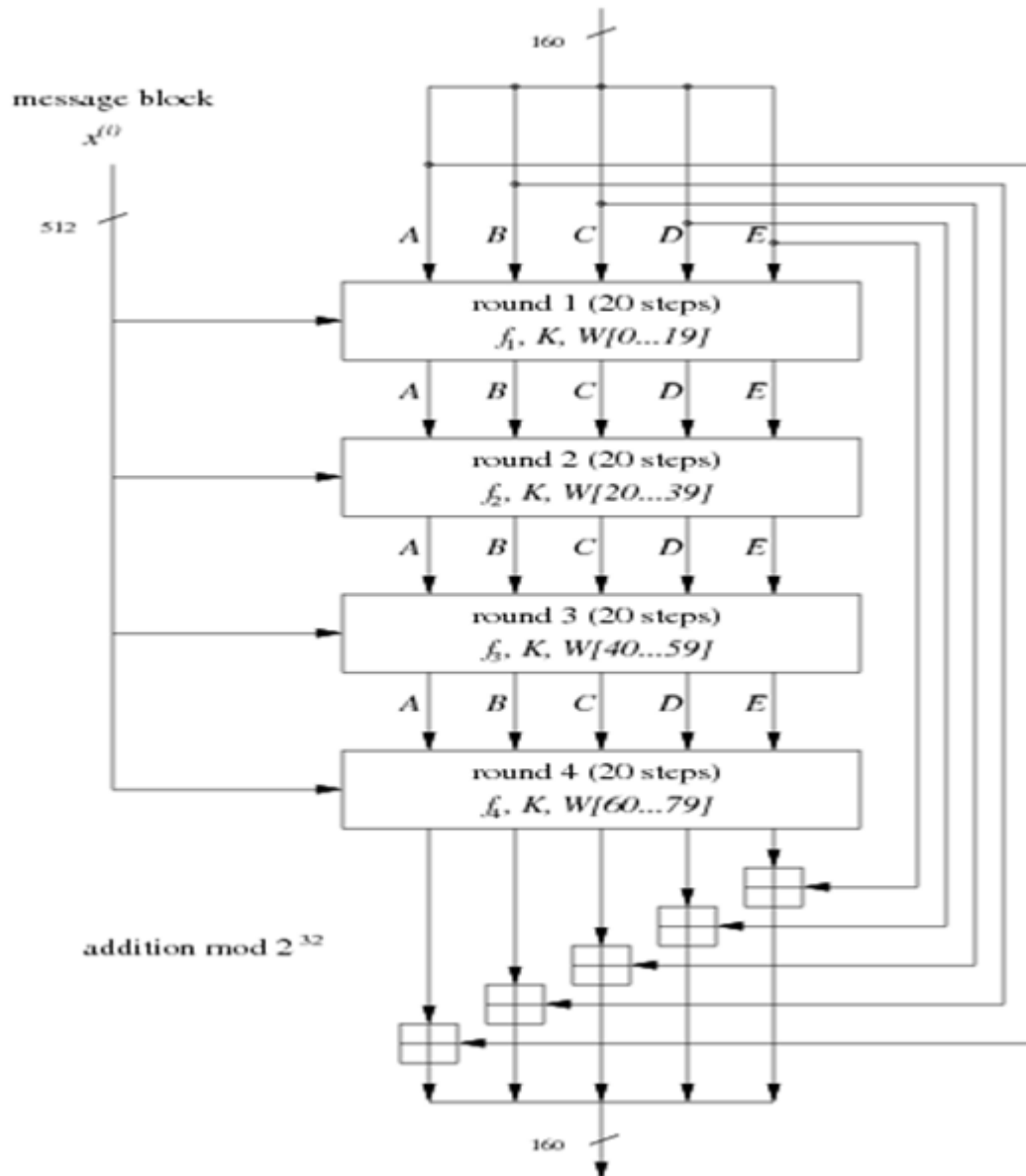
A = H0 = 0x67452301

B = H1 = 0xefcdab89

C = H2 = 0x98badcfe

D = H3 = 0x10325476

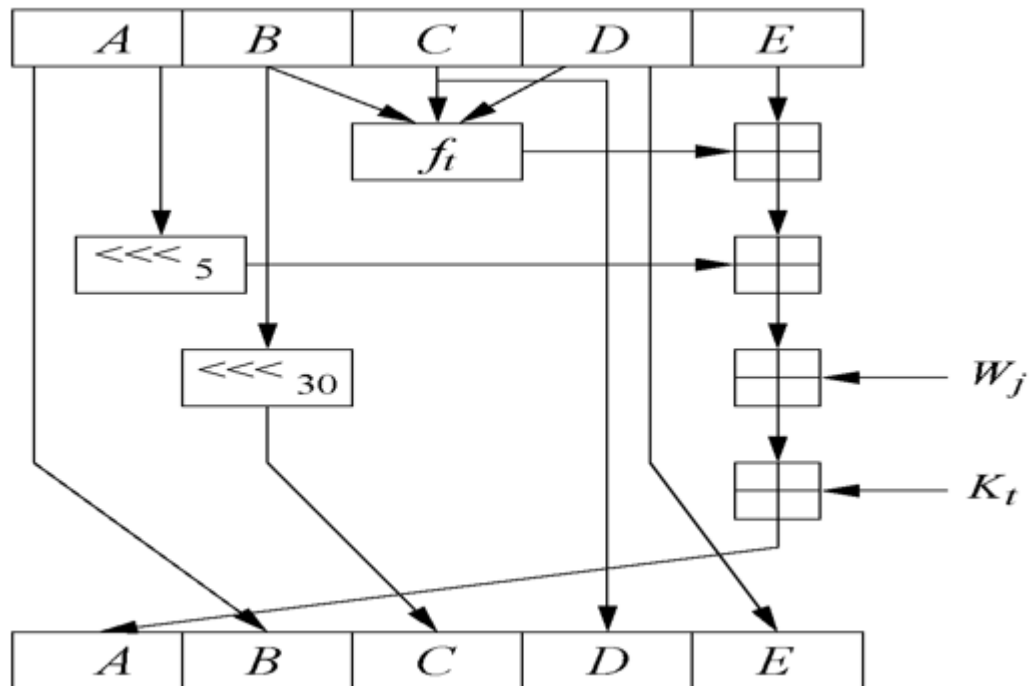
E = H4 = 0xc3d2e1f0



Function: $F(t) = (B \text{ AND } C) \text{ OR } ((\text{NOT } B) \text{ AND } D)$

Constant: $K(t) = 0x5A827999$

Round 0



A0=0x67452301

B0=0xefcdab89

C0=0x98badcfe

D0=0x10325476

E0=0xc3d2e1f0

F Function Calculation:

$$F = (B \text{ AND } C) \text{ OR } ((\text{NOT } B) \text{ AND } D)$$

Input Values:

B = 1110 1111 1100 1101 1010 1011 1000 1001

C = 1001 1000 1011 1010 1101 1100 1111 1110

D = 0001 0000 0011 0010 0101 0100 0111 0110

Step 1: B AND C

B = 1110 1111 1100 1101 1010 1011 1000 1001

C = 1001 1000 1011 1010 1101 1100 1111 1110

B AND C = 1000 1000 1000 1000 1000 1000 1000 1000

Step 2: NOT B

B = 1110 1111 1100 1101 1010 1011 1000 1001

NOT B = 0001 0000 0011 0010 0101 0100 0111 0110

Step 3: (NOT B) AND D

```
NOT B    = 0001 0000 0011 0010 0101 0100 0111 0110
D        = 0001 0000 0011 0010 0101 0100 0111 0110
NOT B & D= 0001 0000 0011 0010 0101 0100 0111 0110
```

Step 4: Final OR operation

```
(B & C)    = 1000 1000 1000 1000 1000 1000 1000 1000
(NOT B & D)= 0001 0000 0011 0010 0101 0100 0111 0110
Final F    = 1001 1000 1011 1010 1101 1100 1111 1110
```

T Value Calculation:

$$T = (A \ll 5) + F + E + W[t] + K[t]$$

Calculate T by rotating A left 5 positions and adding other terms

Step 1: Rotate A left by 5 positions ($A \ll 5$)

```
A          = 0110 0111 0100 0101 0010 0011 0000 0001
A<<<5     = 1110 1000 1010 0100 0110 0000 0010 1100
```

Step 2: Add F function result

```
A<<<5     = 1110 1000 1010 0100 0110 0000 0010 1100
F          = 1001 1000 1011 1010 1101 1100 1111 1110
Sum        = 1000 0001 0101 1111 0011 1101 0010 1010
Note: Modulo  $2^{32}$  arithmetic used
```

Step 3: Add E register value

```
Previous   = 1000 0001 0101 1111 0011 1101 0010 1010
E          = 1100 0011 1101 0010 1110 0001 1111 0000
Sum        = 0100 0101 0011 0010 0001 1111 0001 1010
```

Step 4: Add message word $W[t]$

```
Previous   = 0100 0101 0011 0010 0001 1111 0001 1010
W[t]       = 0100 0001 0110 0010 0110 0100 0111 0101
Sum        = 1000 0110 1001 0100 1000 0011 1000 1111
```

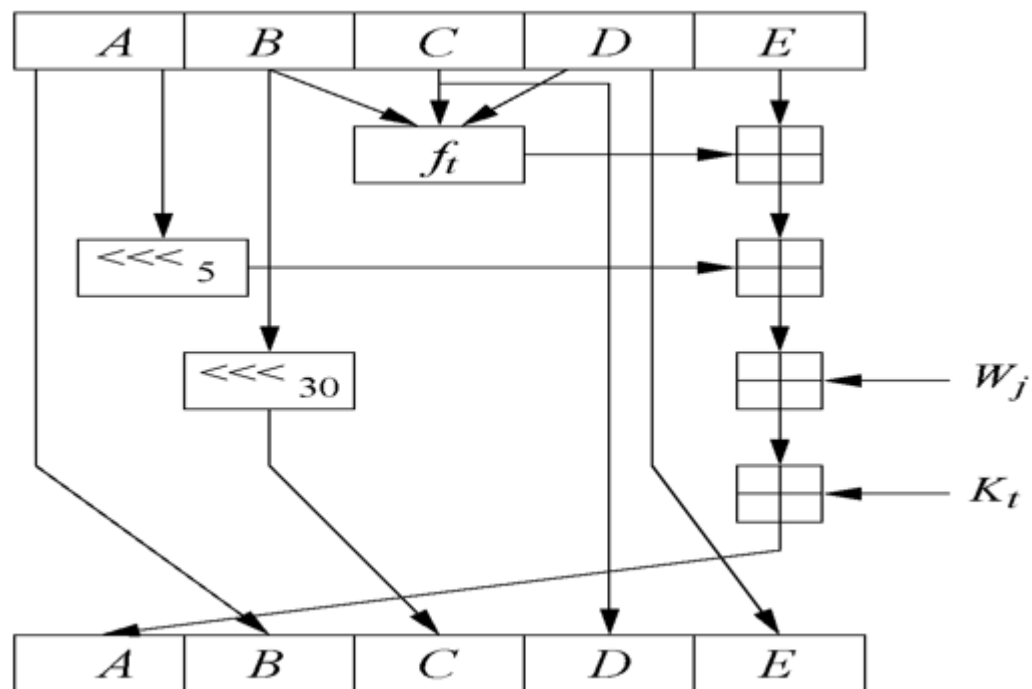
Step 5: Add round constant $K[t]$

```
Previous   = 1000 0110 1001 0100 1000 0011 1000 1111
K[t]       = 0101 1010 1000 0010 0111 1001 1001 1001
Final T    = 1110 0001 0001 0110 1111 1101 0010 1000
```

Register Updates:

```
A1=T=0xe116fd28
B1=A0=0x67452301
C1=B0<<<30=0x7bf36ae2
D1=C0=0x98badcfe
E1=D0=0x10325476
```

Round 1



A1=0xe116fd28

B1=0x67452301

C1=0x7bf36ae2

D1=0x98badcfe

E1=0x10325476

F Function Calculation:

$$F = (B \text{ AND } C) \text{ OR } ((\text{NOT } B) \text{ AND } D)$$

Input Values:

B = 0110 0111 0100 0101 0010 0011 0000 0001

C = 0111 1011 1111 0011 0110 1010 1110 0010

D = 1001 1000 1011 1010 1101 1100 1111 1110

Step 1: B AND C

B = 0110 0111 0100 0101 0010 0011 0000 0001

C = 0111 1011 1111 0011 0110 1010 1110 0010

B AND C = 0110 0011 0100 0001 0010 0010 0000 0000

Step 2: NOT B

B = 0110 0111 0100 0101 0010 0011 0000 0001

NOT B = 1001 1000 1011 1010 1101 1100 1111 1110

Step 3: (NOT B) AND D

```
NOT B    = 1001 1000 1011 1010 1101 1100 1111 1110
D        = 1001 1000 1011 1010 1101 1100 1111 1110
NOT B & D= 1001 1000 1011 1010 1101 1100 1111 1110
```

Step 4: Final OR operation

```
(B & C)    = 0110 0011 0100 0001 0010 0010 0000 0000
(NOT B & D)= 1001 1000 1011 1010 1101 1100 1111 1110
Final F    = 1111 1011 1111 1011 1111 1110 1111 1110
```

T Value Calculation:

$$T = (A \ll 5) + F + E + W[t] + K[t]$$

Calculate T by rotating A left 5 positions and adding other terms

Step 1: Rotate A left by 5 positions ($A \ll 5$)

```
A          = 1110 0001 0001 0110 1111 1101 0010 1000
A << 5     = 0010 0010 1101 1111 1010 0101 0001 1100
```

Step 2: Add F function result

```
A << 5     = 0010 0010 1101 1111 1010 0101 0001 1100
F          = 1111 1011 1111 1011 1111 1110 1111 1110
Sum        = 0001 1110 1101 1011 1010 0100 0001 1010
Note: Modulo  $2^{32}$  arithmetic used
```

Step 3: Add E register value

```
Previous   = 0001 1110 1101 1011 1010 0100 0001 1010
E          = 0001 0000 0011 0010 0101 0100 0111 0110
Sum        = 0010 1111 0000 1101 1111 1000 1001 0000
```

Step 4: Add message word $W[t]$

```
Previous   = 0010 1111 0000 1101 1111 1000 1001 0000
W[t]       = 0110 1100 0110 1100 0110 0001 0110 1000
Sum        = 1001 1011 0111 1010 0101 1001 1111 1000
```

Step 5: Add round constant $K[t]$

```
Previous   = 1001 1011 0111 1010 0101 1001 1111 1000
K[t]       = 0101 1010 1000 0010 0111 1001 1001 1001
Final T    = 1111 0101 1111 1100 1101 0011 1001 0001
```

Register Updates:

```
A2=T=0xf5fcd391
B2=A1=0xe116fd28
C2=B1<<<30=0x59d148c0
D2=C1=0x7bf36ae2
E2=D1=0x98badcfe
```